

Appendix: Mathematical Framework for the Two-Tier Absorbing Markov Chain

To establish a mathematically rigorous projection of the micro-level chess position transitions onto the community-level macro-network, we formalize a two-tier absorbing Markov framework. This framework explicitly resolves the boundary-state transition dynamics between condensed macro-states. Let $D = (V, E)$ represent the global directed position-level chess graph, where $p(u \rightarrow v)$ denotes the probability of transitioning from position u to a neighboring legal position v .

1. Local Sub-Chain and Micro-Canonical Form. For a given transient community $c_i \in \mathcal{T}$, the internal positions $u \in c_i$ are treated as transient states within a localized sub-chain, while all adjacent macro-communities $c_j \in \mathcal{C} \setminus \{c_i\}$ act as distinct absorbing destinations. We construct the local transition matrix $P^{(i)}$ for community c_i in canonical form:

$$P^{(i)} = \begin{bmatrix} Q^{(i)} & R^{(i)} \\ \mathbf{0} & I \end{bmatrix} \quad [7]$$

where the submatrix $Q^{(i)} \in \mathbb{R}^{|c_i| \times |c_i|}$ defines internal position-to-position transitions within the community boundary:

$$Q_{u,v}^{(i)} = \begin{cases} p(u \rightarrow v) & \text{if } v \in c_i \\ 0 & \text{otherwise} \end{cases} \quad [8]$$

and the submatrix $R^{(i)} \in \mathbb{R}^{|c_i| \times (|\mathcal{C}|-1)}$ defines the escaping transition probabilities from internal transient positions directly into adjacent target communities:

$$R_{u,c_j}^{(i)} = \sum_{v \in c_j} p(u \rightarrow v), \quad \forall c_j \neq c_i \quad [9]$$

The local fundamental matrix $N^{(i)} = (I - Q^{(i)})^{-1}$ determines the expected number of times the micro-walk visits each transient position within c_i before escaping the community. The localized absorption probability matrix $B^{(i)}$ is then derived as:

$$B^{(i)} = N^{(i)} R^{(i)} \quad [10]$$

where the matrix element $B_{u,c_j}^{(i)}$ represents the exact probability that a game state currently residing at position $u \in c_i$ will eventually exit into community c_j .

2. State Occupancy Frequencies in Directed Acyclic Graphs.

Because the position-level micro-network $D = (V, E)$ is constructed via a depth-bounded depth-first search (DFS) radiating from a singular initial game state v_{start} , the graph is inherently a finite Directed Acyclic Graph (DAG). Consequently, a conventional steady-state stationary distribution ($\Pi = \Pi P$) is mathematically trivial, concentrating entirely on terminal sinks. To rigorously weight the boundary states, we define a transient state occupancy vector $\mathbf{x} \in \mathbb{R}^{|V|}$, where entry $x(u)$ represents the expected number of times that position u is visited during a game originating at v_{start} .

Let $P_{\text{global}} \in \mathbb{R}^{|V| \times |V|}$ be the transition matrix of the global position graph, and let $\mathbf{e}_{\text{start}} \in \mathbb{R}^{|V|}$ be a standard basis vector that represents a probability mass of 1 at the starting node v_{start} . Because D is a DAG, the matrix P_{global} is strictly

nilpotent, and the global fundamental matrix $\mathcal{N}$ converges analytically via a geometric series:

$$\mathcal{N} = (I - P_{\text{global}})^{-1} = \sum_{k=0}^{\infty} (P_{\text{global}})^k \quad [11]$$

The absolute state occupancy frequencies across the global network are then exactly mapped by:

$$\mathbf{x} = \mathbf{e}_{\text{start}} \mathcal{N} \quad [12]$$

For any transient community $c_i \in \mathcal{T}$, the conditional intra-community state probability $\pi_i(u)$ assigned to position $u \in c_i$ is properly normalized relative to the local community density mass:

$$\pi_i(u) = \frac{x(u)}{\sum_{v \in c_i} x(v)} \quad [13]$$

This formulation ensures that the macro-transition matrix elements P_{ij} accurately reflect the game trajectories that branch directly from the unique starting position of the match.

3. Synthesizing the Macro-Transition Matrix. By taking the expectation of the local absorption probabilities over this conditional distribution, the rigorous macro-transition probability P_{ij} from community c_i to community c_j is defined as:

$$P_{ij} = \sum_{u \in c_i} \pi_i(u) B_{u,c_j}^{(i)} \quad [14]$$

Because $\sum_{j \neq i} B_{u,c_j}^{(i)} = 1$ for all valid positions under guaranteed boundary escape, this formulation strictly satisfies the stochastic row conditions $\sum_{j \neq i} P_{ij} = 1$ and $P_{ii} = 0$, ensuring the mathematical completeness of the macro-network transition matrix P .

4. Boundary Asymptotics and Graph Pruning. The risk-aware edge-weight mapping $w(u, v) = -\ln(1 - P_{\text{defeat}}(v))$ introduces a coordinate singularity when a transient macro-state inevitably leads to a losing absorbing region. If a community v exhibits a global defeat probability of $P_{\text{defeat}}(v) = 1$, the edge cost evaluates to:

$$\lim_{P_{\text{defeat}}(v) \rightarrow 1} [-\ln(1 - P_{\text{defeat}}(v))] = \infty \quad [15]$$

In our hybrid implementation, any directed edge pointing to a community where $P_{\text{defeat}}(v) = 1$ is assigned a cost of ∞ and is structurally pruned from the active adjacency list prior to executing Dijkstra's algorithm.

Conversely, if a destination community v is perfectly secure such that it communicates safely only with the winning absorbing states, its global defeat probability drops to $P_{\text{defeat}}(v) = 0$. The edge cost smoothly yields:

$$w(u, v) = -\ln(1 - 0) = 0 \quad [16]$$

This zero-cost behavior establishes a terminal "sink highway" across the macro-network, naturally drawing Dijkstra's pathfinding algorithm toward structurally verified winning zones with zero accumulated risk penalty.